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IMAGE FORMING SYSTEM AND CHARGE ELIMINATING APPARATUS

Abstract

An image forming system includes: an image forming apparatus configured to form an image on a recording medium; a post-processing apparatus configured to perform post-processing on the recording medium on which an image is formed by the image forming apparatus; a relay apparatus having a function of changing a transport speed of the recording medium on which an image is formed by the image forming apparatus to a transport speed of the post-processing apparatus and transporting the recording medium to a subsequent stage; and a charge eliminating apparatus configured to perform charge elimination on the recording medium that is charged, by reducing a charge amount of the charged recording medium, and the charge eliminating apparatus is provided at a stage that is after the relay apparatus and before the post-processing apparatus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 18/596,937, filed Mar. 6, 2024, which is a continuation application of U.S. patent application Ser. No. 17/405,078 filed Aug. 18, 2021, which claims benefit of Japanese Patent Application No. 2021-066480 filed on Apr. 9, 2021 and Japanese Patent Application No. 2021-066481 filed on Apr. 9, 2021. The disclosure of the prior applications is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an image forming system and a charge elimination apparatus.

2. Related Art

[0003] JP-A-2019-167169 discloses a charge eliminating apparatus including, as charge eliminating units that eliminate charge from a high-resistance charged medium, a first contact charge eliminating unit that eliminates a majority of electric charge charged on the charged medium, the first contact charge eliminating unit including a charge eliminating member that comes into contact with the transported charged medium, and a second non-contact charge eliminating unit that eliminates residual electric charge of the charged medium in a non-contact state after the charge elimination by the first contact charge eliminating apparatus, the second non-contact charge eliminating unit being provided downstream of the first contact charge eliminating unit in the transport direction of the charged medium.

SUMMARY

[0004] Aspects of non-limiting embodiments of the present disclosure relate to providing an image forming system that can uniformly perform charge elimination on a recording medium by a charge eliminating apparatus.

[0005] Aspects of certain non-limiting embodiments of the present disclosure address the above advantages and/or other advantages not described above. However, aspects of the non-limiting embodiments are not required to address the advantages described above, and aspects of the non-limiting embodiments of the present disclosure may not address advantages described above.

[0006] According to an aspect of the present disclosure, there is provided an image forming system including: an image forming apparatus configured to form an image on a recording medium; a post-processing apparatus configured to perform post-processing on the recording medium on

which an image is formed by the image forming apparatus; and a charge eliminating apparatus configured to perform charge elimination on the recording medium that is charged, by reducing a charge amount of the charged recording medium.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:

[0008] FIG. 1 illustrates a configuration of an image forming system **10** according to an exemplary embodiment of the present disclosure;

[0009] FIG. 2 is a perspective view illustrating an external appearance of a charge eliminating apparatus **50** illustrated in FIG. 1;

[0010] FIG. 3 illustrates a structure of the charge eliminating apparatus **50** illustrated in FIG. 2;

[0011] FIG. 4 illustrates details when a charge eliminating roller unit **54** illustrated in FIG. 3 performs charge elimination on a sheet;

[0012] FIG. 5 illustrates an apparatus layout of the image forming system **10** illustrated in FIG. 1;

[0013] FIG. 6 illustrates an apparatus layout of another image forming system;

[0014] FIG. 7 illustrates an apparatus layout of another image forming system;

[0015] FIG. 8 illustrates an apparatus layout of another image forming system;

[0016] FIG. 9 illustrates a reason why a relay apparatus **30** has a transport path longer than a sheet in a transport direction;

[0017] FIG. 10 is a perspective view of the charge eliminating roller unit **54**;

[0018] FIG. 11 illustrates a state in which the charge eliminating roller unit **54** is detached from the body of the charge eliminating apparatus **50**;

[0019] FIG. 12 is a sectional view of the charge eliminating roller unit **54** illustrated in FIG. 10;

[0020] FIG. 13 is an enlarged perspective view of the periphery of a peeling member **63**;

[0021] FIG. 14 illustrates details when the charge eliminating roller unit **54** illustrated in FIG. 3 performs charge elimination on a sheet;

[0022] FIG. 15 illustrates a state in which a sheet winds around a charge eliminating roller **61** when the peeling member **63** is not provided in the charge eliminating roller unit **54**;

[0023] FIG. 16 illustrates a state in which a sheet is prevented from being wound around the charge eliminating roller **61** by the peeling member **63**;

[0024] FIG. 17 illustrates a state in which a sheet is prevented from being wound around the charge eliminating roller **61** by the peeling member **63**;

[0025] FIG. 18 illustrates a state in which charge eliminating rollers **61**, **62** rotate in a forward rotation direction from a stop state;

[0026] FIG. 19 illustrates rotation control when a controller **56** shifts the charge eliminating rollers **61**, **62** from the stop state to a rotation state;

[0027] FIG. 20 is a perspective view of a charge eliminating roller unit **54a** using a plate-shaped peeling member **64**; and

[0028] FIG. 21 is a sectional view of the charge eliminating roller unit **54a** illustrated in FIG. 20.

DETAILED DESCRIPTION

[0029] Next, an exemplary embodiment of the present disclosure will be described in detail with reference to the drawings.

[0030] FIG. 1 illustrates a system configuration of an image forming system **10** according to the exemplary embodiment of the present disclosure.

[0031] As illustrated in FIG. 1, the image forming system **10** according to the exemplary embodiment of the present disclosure includes an image forming apparatus **20**, a relay apparatus

30, an interposer **40**, a charge eliminating apparatus **50**, an inspection apparatus **60**, a stacker **70**, and a post-processing apparatus **80**.

[0032] The image forming apparatus **20** has a function of accommodating a recording medium such as a printing sheet and forming an image on the accommodated printing sheet. The recording medium includes not only a printing sheet but also a film on which an image can be printed. In the following description, the printing sheet (hereinafter, abbreviated as a sheet) will be described as the recording medium.

[0033] The relay apparatus **30** has a function of changing a transport speed of a sheet on which an image is formed by the image forming apparatus **20** to a transport speed of the post-processing apparatus **80** and transporting the sheet to the interposer **40** in a subsequent stage. The relay apparatus **30** according to the exemplary embodiment further has a function of performing post-processing on the sheet on which an image is formed by the image forming apparatus **20**.

[0034] Here, the post-processing function performed by the relay apparatus **30** includes at least one of a sheet cooling function, a curl correction function, a slipsheet insertion function, and a punching function. That is, in the present exemplary embodiment, a post-processing apparatus having such a post-processing function is provided as the relay apparatus **30**. The cooling function is a function of reducing the temperature of a sheet formed with an image. The curl correction function is a function of correcting the curl of the sheet. The slipsheet insertion function is a function of inserting a set slipsheet into the middle of plural sheets formed with images.

[0035] The relay apparatus **30** performs post-processing such as cooling, curl correction, slipsheet insertion, or punching on the sheet on which an image is formed by the image forming apparatus **20**, and transports the sheet to the interposer **40** in a subsequent stage.

[0036] The interposer **40** is a slipsheet insertion apparatus that inserts a set slipsheet in the middle of a transport path.

[0037] The charge eliminating apparatus **50** performs charge elimination of reducing a charge amount of a charged sheet. A specific structure for eliminating the charge from the sheet by the charge eliminating apparatus **50** will be described later.

[0038] The post-processing apparatus **80** performs post-processing on the sheet on which an image is formed by the image forming apparatus **20**. Specifically, the post-processing apparatus **80** has one or plural functions of a creasing function, a cutting function, a bookbinding function, a binding function, a stacking function, an alignment function, a punching function, and a sheet discharge function for the sheet on which an image is formed by the image forming apparatus **20**. Here, the creasing function is a function of creasing a sheet. The cutting function is a function of cutting tops and bottoms and fore edges of sheets formed as a booklet. The bookbinding function is also referred to as a booklet function and is a function of bundling plural sheets into a booklet. The binding function is also referred to as a stapling function and is a function of binding plural sheets using a staple. The stacking function is a function of sequentially stacking transported sheets on a tray. The alignment function is also referred to as a tamper/jogger fence function and is a function of aligning a sheet in a width direction or the like. The punching function is a function of forming a punched hole in a predetermined position of a sheet.

[0039] The image forming system **10** according to the present exemplary embodiment further includes the interposer **40** that is a slipsheet insertion apparatus that inserts a set slipsheet into the middle of the transport path.

[0040] The image forming system **10** according to the present exemplary embodiment further includes the inspection apparatus **60** that inspects whether an image is normally formed on a sheet.

[0041] The inspection apparatus **60** receives the sheet on which an image is printed from the image forming apparatus **20** from a preceding stage, and performs inspection of the image printed on the sheet, that is, determines whether there is an abnormality in the image printed on the sheet.

Specifically, the inspection apparatus **60** receives image data of the image printed on the sheet from the image forming apparatus **20**, reads the image on the sheet transported from the image forming

apparatus **20**, collates the image of the received image data with the read image, and determines whether the image printed on the transported sheet is abnormal.

[0042] The stacker **70** accommodates the sheet on which an image is printed by the image forming apparatus **20** therein when the sheet is transported to the stacker **70**.

[0043] Since the image forming system **10** according to the present exemplary embodiment is configured as described above, the sheet on which an image is printed in the image forming apparatus **20** is transported in an order of the relay apparatus **30**, the interposer **40**, the charge eliminating apparatus **50**, the inspection apparatus **60**, the stacker **70**, and the post-processing apparatus **80**, and various processes are performed.

[0044] Here, the image forming apparatus **20** performs printing in which an electrostatic latent image is formed by irradiating a charged photoreceptor with a laser beam or the like, the formed electrostatic latent image is developed with a colorant such as a toner, the developed image is transferred to a sheet, and then fixing processing is performed. For this reason, a sheet or the like on which an image is formed in the image forming apparatus **20** is usually charged. In particular, when a high-resistance dielectric such as a resin film is used as the recording medium, the electric charge charged on the recording medium may increase. When a front surface of the recording medium is charged with, for example, negative electric charge having a negative polarity, positive electric charge having a positive polarity is induced on a back surface of the recording medium by dielectric polarization. As a result, when recording media are attracted to each other by electrostatic attraction and post-processing is performed in the post-processing apparatus **80** in a state where the recording media are charged, the recording media may overlap each other on a sheet discharge tray and are thus charged and adhered to each other, or may discharge charge to a human body.

[0045] Further, when the recording medium such as a charged sheet passes through the inspection apparatus **60**, noise may be generated at the time of inspection and the inspection result may be incorrect.

[0046] For this reason, in the image forming system **10** according to the present exemplary embodiment, the charge eliminating apparatus **50** that eliminates charge of reducing the electric charge charged on the recording medium is provided at a subsequent stage of the image forming apparatus **20**.

[0047] An external appearance of the charge eliminating apparatus **50** illustrated in FIG. **1** will be described with reference to a perspective view of FIG. **2**. As illustrated in FIG. **2**, the charge eliminating apparatus **50** is connectable to other apparatuses at a preceding stage and a subsequent stage, receives a sheet transported from the apparatus at the preceding stage, performs charge elimination processing, and then transports the sheet to the apparatus at the subsequent stage. Since the image forming system **10** according to the present exemplary embodiment is configured as illustrated in FIG. **1**, the charge eliminating apparatus **50** receives the sheet transported from the interposer **40** at the preceding stage, performs the charge elimination processing, and then transports the sheet after charge elimination to the inspection apparatus **60** at the subsequent stage.

[0048] Next, the structure of the charge eliminating apparatus **50** illustrated in FIG. **2** will be described with reference to FIG. **3**. FIG. **3** is a perspective view of the inside of the charge eliminating apparatus **50** illustrated in FIG. **2** when viewed from the front.

[0049] As illustrated in FIG. **3**, the charge eliminating apparatus **50** is configured to transport a sheet transported from an apparatus at the preceding stage to an apparatus at the subsequent stage by three pairs of transport rollers **51** to **53**. A charge eliminating roller unit **54** is provided between the pair of transport rollers **51** and the pair of transport rollers **52**, and a non-contact charge eliminating unit **55** is provided between the pair of transport rollers **52** and the pair of transport rollers **53**.

[0050] The charge eliminating roller unit **54** includes two charge eliminating rollers **61**, **62**. The two charge eliminating rollers **61**, **62** sandwich and transport a sheet to perform charge elimination. The charge eliminating roller unit **54** is detachable from the body of the charge eliminating

apparatus **50**.

[0051] Details of the charge elimination of the sheet by the charge eliminating roller unit **54** illustrated in FIG. **3** will be described with reference to FIG. **4**.

[0052] As illustrated in FIG. **4**, the charge eliminating rollers **61**, **62** constitute a pair, a driving force from a driving motor is transmitted to the charge eliminating roller **62**, and the charge eliminating roller **61** is brought into contact with the charge eliminating roller **62** to be driven by the charge eliminating roller **62**.

[0053] A charge eliminating power supply is connected to the charge eliminating roller **62**, and a charge eliminating bias V_a , for example, is applied from the charge eliminating power supply to the charge eliminating roller **62**. In the present exemplary embodiment, a positive DC voltage is used as the charge eliminating bias V_a . The charge eliminating roller **61** is grounded.

[0054] In the charge eliminating roller unit **54**, when the charge eliminating bias V_a is applied to the charge eliminating roller **62**, the charge eliminating roller unit **54** applies positive electric charge to the front surface of the charged sheet as the sheet passes through. As a result, the nonpolar negative electric charge on the front surface of the sheet is eliminated by an amount of electric charge applied to the sheet. As the electric charge on the front surface of the sheet decreases, the positive electric charge having the positive polarity that is dielectric-polarized on the back surface of the sheet also decreases.

[0055] In this way, the charge eliminating roller unit **54**, which is a contact charge eliminator, functions as a contact charge eliminator and comes into direct contact with the sheet to apply electric charge thereto, so that a large amount of charge can be eliminated to some extent. However, a variation amount of a surface potential of the sheet after charge elimination is large, and the charge elimination tends to be not uniform.

[0056] The non-contact charge eliminating unit **55** is, for example, a corotron type charge eliminating device or a scorotron type charge eliminating device, and has a function of eliminating electric charge without coming into contact with the sheet by supplying the sheet with ions generated by performing corona discharge using, for example, an AC voltage. With the non-contact charge eliminating unit **55**, although a large amount of charge elimination cannot be ensured, the variation amount of the surface potential after charge elimination is small and the charge elimination can be uniformly performed.

[0057] In the charge eliminating apparatus **50** according to the present exemplary embodiment, the charge eliminating roller unit **54** and the non-contact charge eliminating unit **55**, which are two charge eliminators having different characteristics, perform charge elimination on the sheet transported from an apparatus at the preceding stage, and transport the sheet to an apparatus at the subsequent stage.

[0058] However, in any of the charge eliminators of the charge eliminating roller unit **54** and the non-contact charge eliminating unit **55**, an amount of electric charge to be eliminated varies depending on time for passage of the sheet. That is, the amount of electric charge to be eliminated increases as the time for passage of the sheet increases, and the amount of electric charge to be eliminated decreases as the time for passage of the sheet decreases.

[0059] Here, the transport speed of the sheet or the like in the image forming apparatus **20** is slower than the transport speed of the sheet or the like in the post-processing apparatus **80**. For this reason, the sheet on which an image is formed in the image forming apparatus **20** is transported to the post-processing apparatus **80** after the transport speed is increased.

[0060] For this reason, if the charge eliminating apparatus **50** is connected immediately after the image forming apparatus **20**, the transport speed increases during the charge elimination in the charge eliminating apparatus **50**, and the charge elimination would not be uniformly performed.

[0061] Therefore, in the image forming system **10** according to the present exemplary embodiment, the relay apparatus **30** is provided between the charge eliminating apparatus **50** and the image forming apparatus **20**.

[0062] That is, in the image forming system **10** according to the present exemplary embodiment, the charge eliminating apparatus **50** is provided at a stage after the relay apparatus **30** and a stage before the post-processing apparatus **80**.

[0063] As described above, the relay apparatus **30** has a function of changing the transport speed of the sheet on which an image is formed by the image forming device **20** to the transport speed of the post-processing apparatus **80** and transporting the sheet to the interposer **40** at the subsequent stage. For this reason, what is transported to the charge eliminating apparatus **50** is the sheet having the transport speed increased after passing through the relay apparatus **30**. The relay apparatus **30** has a transport path longer than the sheet in the transport direction. A reason why the length of the transport path in the relay apparatus **30** is set in this manner will be described later.

[0064] In the image forming system **10** according to the present exemplary embodiment, the interposer **40** is provided in the middle of the transport path of the sheet, so that a slipsheet can be inserted in the middle of the transport path.

[0065] However, the slipsheet is usually used in a state in which characters are printed, and the slipsheet may be set in the interposer **40** in a charged state. The slipsheet may also be charged when inserted by the interposer **40**. When the interposer **40** is provided at a stage after the charge eliminating apparatus **50**, a sheet in a charged state would be transported to the post-processing apparatus **80**.

[0066] For this reason, in the present exemplary embodiment, the charge eliminating apparatus **50** is provided at a stage after the interposer **40**.

[0067] As described above, when the sheet passes through the inspection apparatus **60** in a charged state, noise may be generated at the time of inspection in the inspection apparatus **60** and the inspection result may be incorrect. For this reason, the charge eliminating apparatus **50** is provided at a stage before the inspection apparatus **60**.

[0068] FIG. **5** illustrates an apparatus layout of the image forming system **10** according to the present exemplary embodiment described above.

[0069] As illustrated in FIG. **5**, the relay apparatus **30** is provided between the charge eliminating apparatus **50** and the image forming apparatus **20**, and the transport speed of the sheet transported to the charge eliminating apparatus **50** has already been increased. The charge eliminating apparatus **50** is provided at a stage after the interposer **40** and a stage before the inspection apparatus **60**.

[0070] FIG. **6** illustrates an apparatus layout of an image forming system in which the interposer **40** and the stacker **70** are not provided. As illustrated in FIG. **6**, even in the case of the apparatus layout in which the interposer **40** and the stacker **70** are not provided, the above-described problem can also be prevented if the charge eliminating apparatus **50** is provided at a stage after the relay apparatus **30** and the relay apparatus **30** is provided between the charge eliminating apparatus **50** and the image forming apparatus **20**.

[0071] FIG. **7** illustrates an apparatus layout of an image forming system in which the inspection apparatus **60** is not provided. As illustrated in FIG. **7**, even in the case of the apparatus layout in which the inspection apparatus **60** is not provided, the above-described problem can also be prevented if the charge eliminating apparatus **50** is provided at a stage after the relay apparatus **30** and a stage before the post-processing apparatus **80**.

[0072] FIG. **8** illustrates an apparatus layout of an image forming system in which a discharge tray is attached to the charge eliminating apparatus **50** and the post-processing apparatus **80** is not provided.

[0073] FIG. **8** illustrates a case where the charge eliminating apparatus **50** is used in a state where the discharge tray is attached thereto, and the post-processing apparatus **80** is not provided. Even in the case of such an apparatus layout, when the charge eliminating apparatus **50** is provided at a stage after the relay apparatus **30**, the sheet after the transport speed is changed is fed to the charge eliminating apparatus **50**.

[0074] Finally, the reason why the relay apparatus **30** has a transport path longer than the sheet in the transport direction will be described with reference to FIG. **9**. For example, a most frequently used A4-size sheet has a length of 210 mm in the transport direction. For this reason, by setting the horizontal width of the relay apparatus **30**, that is, the length of the transport path to 210 mm or more, it is possible to ensure that the sheet after passing through the relay apparatus **30** is transported to an apparatus at the subsequent stage in a state in which the transport speed of the sheet is changed.

[0075] For this reason, even when the charge eliminating apparatus **50** is connected immediately after the relay apparatus **30**, the transport speed of the sheet transported to the charge eliminating apparatus **50** is a transport speed after change, and is prevented from being changed in the middle of the charge elimination.

[0076] The charge eliminating apparatus **50** according to the present exemplary embodiment will be described in detail.

[0077] In the charge eliminating apparatus **50** according to the present exemplary embodiment, the charge eliminating roller unit **54** and the non-contact charge eliminating unit **55**, which are two charge eliminators having different characteristics, perform charge elimination on the sheet transported from an apparatus at the subsequent stage, and transport the sheet to an apparatus at the subsequent stage.

[0078] Then, a controller **56** controls operations of the three pairs of transport rollers **51** to **53**, the charge eliminating roller unit **54**, and the non-contact charge eliminating unit **55** to perform charge elimination processing on the sheet.

[0079] As described above, the charge eliminating roller unit **54** includes the two charge eliminating rollers **61**, **62**, and the controller **56** controls rotation of the two charge eliminating rollers **61**, **62**.

[0080] The sheet is sandwiched and transported by the two charge eliminating rollers **61**, **62** to perform charge elimination. The charge eliminating roller unit **54** is detachable from the body of the charge eliminating apparatus **50**.

[0081] A perspective view of the charge eliminating roller unit **54** is illustrated in FIG. **10**. As illustrated in FIG. **10**, in the charge eliminating roller unit **54**, the two charge eliminating rollers **61**, **62** are respectively provided on lower and upper sides in a vertical direction, and plural peeling members **63** are abutted against the charge eliminating roller **61** provided on the lower side in the vertical direction.

[0082] The peeling members **63** are arranged side by side in the width direction of the transport path along which the sheet is transported.

[0083] In this way, the two charge eliminating rollers **61**, **62** and the plural peeling members **63** are provided in the charge eliminating roller unit that is insertable into and removable from the body of the charge eliminating apparatus **50**.

[0084] FIG. **11** illustrates a state in which the charge eliminating roller unit **54** is detached from the body of the charge eliminating apparatus **50**. As illustrated in FIG. **11**, the charge eliminating apparatus **50** includes a guide plate **57** that guides an insertion position of the charge eliminating roller unit **54** when the charge eliminating roller unit **54** is inserted into the body of the charge eliminating apparatus **50**. A part of the guide plate **57** protrudes from an attachment opening for attaching the charge eliminating roller unit **54**. For this reason, when the charge eliminating roller unit **54** is inserted into the charge eliminating apparatus **50**, a user can easily attach the charge eliminating roller unit **54** to the body of the charge eliminating apparatus **50** by placing a lower portion of the charge eliminating roller unit **54** on the guide plate **57**.

[0085] Next, a cross-sectional view of the charge eliminating roller unit **54** illustrated in FIG. **10** is illustrated in FIG. **12**.

[0086] The charge eliminating roller unit **54** includes the two charge eliminating rollers **61**, **62** and the peeling members **63**. The two charge eliminating rollers **61**, **62** are applied with voltages of

different potentials and perform charge elimination by reducing the charge amount of the sheet when the sheet passes through the charge eliminating rollers **61**, **62**.

[0087] The peeling members **63** are abutted against a surface of the charge eliminating roller **61** of the two charge eliminating rollers **61**, **62**, and peel off the sheet adhering to the charge eliminating roller **61** with the rotation of the charge eliminating roller **61**.

[0088] Each of the peeling members **63** includes a movable portion **71**, a support portion **72**, and a spring **73** that is an elastic body.

[0089] The support portion **72** is fixed to a housing of the charge eliminating roller unit **54** and supports the movable portion **71** to be movable. The movable portion **71** is movably supported by the support portion **72** and is biased toward the surface of the charge eliminating roller **61** by the spring **73**. The movable portion **71** is abutted against the surface of the charge eliminating roller **61** by being biased in direction toward the surface of the charge eliminating roller **61** by the spring **73**.

[0090] A top end portion of the movable portion **71**, which is a top end of the peeling member **63**, is abutted against the downstream surface of the charge eliminating roller **61** in a forward rotation direction at an acute angle. The peeling member **63** is provided in a position where the peeling member **63** peels off the sheet adhering to the surface of the charge eliminating roller **61** with the rotation of the charge eliminating roller **61**.

[0091] With the above-described configuration, the peeling member **63** functions to peel off the sheet adhering to the charge eliminating roller **61** with the rotation of the charge eliminating roller **61**.

[0092] FIG. **13** is an enlarged perspective view of the periphery of the peeling member **63**.

Referring to FIG. **13**, it can be seen that the movable portion **71** rotatably attached to the support portion **72** is biased by the spring **73**.

[0093] Next, details of the charge elimination of the sheet by the charge eliminating roller unit **54** illustrated in FIG. **3** will be described with reference to FIG. **14**. In FIG. **14**, only the movable portion **71** of the peeling member **63** is schematically illustrated instead of illustrating the entire peeling member **63**.

[0094] As illustrated in FIG. **14**, the charge eliminating rollers **61**, **62** constitute a pair, a driving force from a driving motor is transmitted to the charge eliminating roller **62**, and the charge eliminating roller **61** is brought into contact with the charge eliminating roller **62** to be driven by the charge eliminating roller **62**.

[0095] A charge eliminating power supply is connected to the charge eliminating roller **62**, and the charge eliminating bias V_a , for example, is applied from the charge eliminating power supply to the charge eliminating roller **62**. In the present exemplary embodiment, a positive DC voltage is used as the charge eliminating bias V_a . The other charge eliminating roller **61** is grounded.

[0096] In the charge eliminating roller unit **54**, when the charge eliminating bias V_a is applied to the charge eliminating roller **62**, the charge eliminating roller unit **54** applies positive electric charge to the front surface of the charged sheet as the sheet passes through. As a result, the nonpolar negative electric charge on the front surface of the sheet is eliminated by an amount of electric charge applied to the sheet. As the electric charge on the front surface of the sheet decreases, the positive electric charge having the positive polarity that is dielectric-polarized on the back surface of the sheet also decreases.

[0097] A top end of the movable portion **71** constituting a part of the peeling member **63** is abutted against the surface of the charge eliminating roller **61**. The peeling member **63** functions to peel the sheet adhering to the charge eliminating roller **61** with the rotation of the charge eliminating roller **61**.

[0098] The peeling member **63** is provided to prevent sheet jam due to winding of the sheet after charge elimination around the charge eliminating roller **61** on a ground side. FIG. **15** illustrates a state in which the sheet winds around the charge eliminating roller **61** when the peeling member **63**

is not provided on the charge eliminating roller unit **54**.

[0099] FIG. **15** illustrates a state in which, when charge elimination is performed on a sheet **S**, the sheet **S** is charged due to positive electric charge applied to the sheet from the charge eliminating roller **62** having a positive potential to which the charge eliminating bias V_a is applied, and the sheet **S** is wound around the charge eliminating roller **61** due to an increase in a potential difference between the sheet **S** and the charge eliminating roller **61** on the ground side that serves as a counter electrode. Such winding around the charge eliminating roller **61** is likely to occur when charge elimination is performed on a so-called low-stiffness sheet, for example, a thin sheet.

[0100] When the sheet **S** winds around the charge eliminating roller **61** in this way, the sheet **S** after charge elimination would not reach the pair of transport rollers **52** and the sheet jam occurs.

[0101] Next, a state in which the sheet is prevented from being wound around the charge eliminating roller **61** by the peeling member **63** will be described with reference to FIGS. **16** and **17**.

[0102] In the case where the peeling member **63** is provided as illustrated in FIG. **16**, the top end portion of the movable portion **71** of the peeling member **63** is in direct contact with the surface of the charge eliminating roller **61**. The top end portion of the movable portion **71** defines an acute angle with the downstream surface of the charge eliminating roller **61**. For this reason, as illustrated in FIG. **16**, even when the sheet **S** that has passed through between the two charge eliminating rollers **61**, **62** adheres to the charge eliminating roller **61**, a leading end of the sheet **S** is separated from the surface of the charge eliminating roller **61** by the movable portion **71**.

[0103] As a result, as illustrated in FIG. **17**, the sheet **S** reaches the pair of transport rollers **52** normally and is transported to an apparatus at the subsequent stage without sheet jam.

[0104] Next, rotation control of the charge eliminating rollers **61**, **62** will be described. As described above, in the charge eliminating apparatus **50**, the rotation of the charge eliminating rollers **61**, **62** is controlled by the controller **56** illustrated in FIG. **3**.

[0105] The controller **56** controls, before controlling the two charge eliminating rollers **61**, **62** to perform forward rotation operation of rotating in the forward rotation direction in which the sheet passes through between the two charge eliminating rollers **61**, **62** and is transported to the subsequent stage, the charge eliminating roller **61** provided with the peeling member **63** to perform reverse rotation operation of rotating in the reverse rotation direction opposite to the forward rotation direction.

[0106] The controller **56** also controls the charge eliminating roller **61** to perform the reverse rotation operation not only at the start of a printing processing on the sheet but also after the printing processing on the sheet based on a print job that is a print instruction.

[0107] At this time, the controller **56** controls the charge eliminating roller **61** to temporarily stop after the reverse rotation operation, and to then perform the forward rotation operation.

[0108] When the charge eliminating roller **61** performs the forward rotation operation after the reverse rotation operation, the controller **56** performs control to gradually increase the rotation speed of the charge eliminating roller **61** during the forward rotation operation until steady rotation.

[0109] A reason why the controller **56** does not control the two charge eliminating rollers **61**, **62** to perform the forward rotation operation from a stop state when the controller **56** performs charge elimination on the sheet in this way will be described with reference to FIG. **18**.

[0110] FIG. **18** illustrates a state in which the charge eliminating rollers **61**, **62** rotate in the forward rotation direction from the stop state.

[0111] In the charge eliminating apparatus **50** according to the exemplary embodiment, the top end of the movable portion **71** constituting the peeling member **63** is abutted against the charge eliminating roller **61** in a state of being biased in the direction toward the surface of the charge eliminating roller **61**. For this reason, when printing is not performed for a long period of time in the image forming system **10**, the top end of the movable portion **71** and the surface of the charge eliminating roller **61** continue being in contact with each other. In the charge eliminating apparatus

50, a discharge product such as nitrogen oxide due to corona discharge may be generated.

[0112] When the top end of the movable portion **71** continues being in contact with the surface of the charge eliminating roller **61** for a long time in presence of such a discharge product, the top end of the movable portion **71**, which is a part of the peeling member **63**, may be bonded (or fixed) to the surface of the charge eliminating roller **61** due to the influence of the discharge product. [0113] (A) of FIG. **18** illustrates a state in which the top end of the movable portion **71** is bonded to the surface of the charge eliminating roller **61** in this way. Referring to (A) of FIG. **18**, it can be seen that the top end of the movable portion **71** biased by the spring **73** is bonded to the surface of the charge eliminating roller **61**. In FIG. **18**, to facilitate understanding of the description, a mark is provided in a position on the surface of the charge eliminating roller **61** to which the movable portion **71** is bonded. [0114] (B) of FIG. **18** illustrates a state in which the charge eliminating rollers **61**, **62** are rotated in the forward rotation direction in such a state. In (B) of FIG. **18**, it can be seen that the top end portion of the movable portion **71** is bent due to the charge eliminating roller **61** rotating in the forward rotation direction in a state where the top end portion of the movable portion **71** is bonded to the surface of the charge eliminating roller **61**. When the charge elimination processing is performed in a state in which the top end portion of the movable portion **71** is bent in this way, the peeling member **63** would not function normally and the sheet jam occurs.

[0115] Therefore, in the charge eliminating apparatus **50** according to the present exemplary embodiment, the controller **56** performs rotation control by a method as illustrated in FIG. **19** when the charge eliminating rollers **61**, **62** are shifted from the stop state to a rotation state.

[0116] First, when the two charge eliminating rollers **61**, **62** in the stop state as illustrated in (A) of FIG. **19** are to be rotated, the controller **56** rotates the charge eliminating rollers **61**, **62** in the reverse direction opposite to the forward direction as illustrated in (B) of FIG. **19**. When doing so, the bonding between the movable portion **71** and the surface of the charge eliminating roller **61** is released. Thereafter, the controller **56** rotates the charge eliminating rollers **61**, **62** in the forward rotation direction in which the charge eliminating rollers **61**, **62** are originally intended to be rotated. When doing so, as illustrated in (C) FIG. **19**, the top end of the movable portion **71** maintains being normally abutted against the surface of the charge eliminating roller **61** without being bent.

[0117] In this way, the controller **56** performs control, before rotating the two charge eliminating rollers **61**, **62** in the forward rotation direction, to rotate the two charge eliminating rollers **61**, **62** in the forward rotation direction after controlling the two charge eliminating rollers **61**, **62** to perform the reverse rotation operation of rotating in the reverse rotation direction opposite to the forward rotation direction.

[0118] Here, in the charge eliminating roller unit **54** described above, plural peeling members **63** constituted by the movable portion **71** and the support portion **72** are provided, and the peeling members may have other configurations.

[0119] For example, FIG. **20** illustrates a charge eliminating roller unit **54a** using one plate-shaped peeling member **64**. The peeling member **64** illustrated in FIG. **20** is formed of, for example, a mylar film, and has a top end portion abutted against the surface of the charge eliminating roller **61** similarly to the peeling members **63**.

[0120] FIG. **21** is a cross-sectional view of the charge eliminating roller unit **54a** illustrated in FIG. **20**. Referring to FIG. **21**, it can be seen that the plate-shaped peeling member **64** is provided to the charge eliminating roller **61** at an angle at which the transported sheet is to be peeled off. Even with such a plate-shaped peeling member **64**, similar to the peeling members **63** described above, the sheet attached to the surface of the charge eliminating roller **61** is peeled off.

[0121] The foregoing description of the exemplary embodiments of the present invention has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Obviously, many modifications and variations

will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, thereby enabling others skilled in the art to understand the invention for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention defined by the following claims and their equivalents.

REFERENCE SIGN LIST

[0122] **10** image forming system [0123] **20** image forming apparatus [0124] **30** relay apparatus [0125] **40** interposer [0126] **50** charge eliminating apparatus [0127] **51** to **53** pair of transport roller [0128] **54**, **54a** charge eliminating roller unit [0129] **55** non-contact charge eliminating unit [0130] **56** controller [0131] **57** guide plate [0132] **60** inspection apparatus [0133] **61**, **62** charge eliminating roller [0134] **63** peeling member [0135] **64** peeling member [0136] **70** stacker [0137] **70** stacker [0138] **71** movable portion [0139] **72** support portion [0140] **73** spring [0141] **80** post-processing apparatus

Claims

1. An image forming system comprising: an image forming member configured to form an image on a recording medium, the image forming member being configured to perform: printing in which a formed electrostatic latent image is developed with a toner, and transferring the developed image to the recording medium; a post-processing member configured to perform post-processing on the recording medium on which the image is formed by the image forming member; a fixing member configured to perform fixing processing after the image forming member performs transferring the developed image to the recording medium, a charge eliminating member configured to perform charge elimination on the recording medium that is charged by reducing a charge amount of the charged recording medium, the charge eliminating member being positioned at a stage after the image forming member and before the post-processing member in a processing direction, a sheet cooling function member and a curl correction member being positioned downstream of the fixing member, and an interposer positioned downstream of the sheet cooling function member and the curl correction member and upstream of the charge eliminating member, wherein the charge eliminating member includes: a non-contact charge eliminator that generates ions, and has a function of eliminating electric charge without coming into contact with the recording medium by supplying the recording medium with the ions generated by the non-contact charge eliminator, the non-contact charge eliminator generating the ions by performing corona discharge using an AC voltage, and a contact charge eliminator, the contact charge eliminator being two charge eliminating rollers configured to: apply positive electric charge to a front surface of the recording medium, and sandwich and transport the recording medium to perform the charge elimination, the non-contact charge eliminator is positioned downstream of the contact charge eliminator.
2. The image forming system according to claim 1, wherein an inspection member is positioned downstream of the charge eliminating member.
3. The image forming system according to claim 1, wherein the non-contact charge eliminator is a corotron type charge eliminating device or a scorotron type charge eliminating device.
4. The image forming system according to claim 1, wherein the sheet cooling function member or the curl correction member is a relay member, and a transport speed of the recording medium is increased after passing through the relay member.
5. The image forming system according to claim 1, wherein the post-processing member has a bookbinding function.
6. An image forming system comprising: an image forming unit configured to form an image on a recording medium, the image forming unit being configured to perform: printing in which a formed electrostatic latent image is developed with a toner, and transferring the developed image to the recording medium; a post-processing unit configured to perform post-processing on the recording

medium on which the image is formed by the image forming unit; a fixing unit configured to perform fixing processing after the image forming unit performs transferring the developed image to the recording medium, a charge eliminating unit configured to perform charge elimination on the recording medium that is charged by reducing a charge amount of the charged recording medium, the charge eliminating unit being positioned at a stage after the image forming unit and before the post-processing unit in a processing direction, a sheet cooling function unit and a curl correction unit being positioned downstream of the fixing unit, and an interposer positioned downstream of the sheet cooling function unit and the curl correction unit and upstream of the charge eliminating unit, wherein the charge eliminating unit includes: a non-contact charge eliminator that generates ions, and has a function of eliminating electric charge without coming into contact with the recording medium by supplying the recording medium with the ions generated by the non-contact charge eliminator, the non-contact charge eliminator generating the ions by performing corona discharge using an AC voltage and, a contact charge eliminator, the contact charge eliminator being two charge eliminating rollers configured to: apply positive electric charge to a front surface of the recording medium, and sandwich and transport the recording medium to perform the charge elimination, the non-contact charge eliminator is positioned downstream of the contact charge eliminator.

7. The image forming system according to claim 6, wherein an inspection unit is positioned downstream of the charge eliminating unit.

8. The image forming system according to claim 6, wherein the non-contact charge eliminator is a corotron type charge eliminating device or a scorotron type charge eliminating device.

9. The image forming system according to claim 6, wherein the sheet cooling function unit or the curl correction unit is a relay unit, and a transport speed of the recording medium is increased after passing through the relay unit.

10. The image forming system according to claim 6, wherein the post-processing unit has a bookbinding function.
